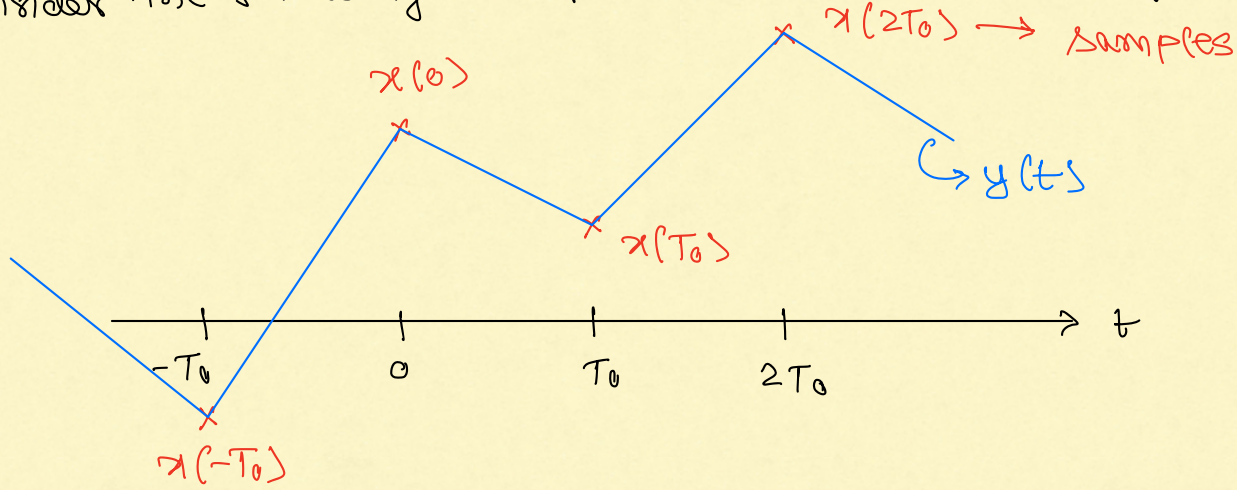
Practice Set 9

① Reconstruction via linear interpolation:

Assume that a signal $x(t)$ is sampled to obtain $x(nT_0)$, $n \in \mathbb{Z}$.

Consider the following simple reconstruction technique:



This is called linear interpolation. Of course, there is no guarantee that linear interpolation can actually reconstruct $x(t)$ accurately; however, this can be used as an approximation.

(a) Let $y(t)$ denote the signal obtained via linear interpolation.

Find a signal $h(t)$ such that

$$y(t) = \underbrace{\left[\sum_n x(nT_0) \cdot \delta(t - nT_0) \right]}_{\text{sampled signal}} * \underbrace{h(t)}_{\text{impulse response of a filter.}}$$

(b) Assuming that the convolution property can be applied, find $Y(\omega)$.

(c) When can we say that $Y(\omega)$ is a reasonable approximation for $X(\omega)$?

(2) <u>Another viewpoint</u> <u>for sampling theorem</u>:

Suppose $x(t)$ is bandlimited to $-\omega_{max} \leq \omega \leq +\omega_{max}$ and sampling frequency $\omega_0 > 2\omega_{max}$.

Let $z(t) = x(t) \cdot \sum_n \delta(t - nT_0)$.

(a) Is $Z(\omega)$ periodic as a function of ω? What is its period?

(b) Since $Z(\omega)$ is a periodic function, we can defined a Fourier series representation of Z by treating ω as how we would usually treat time.

Find the value of the nth Fourier series coefficient of $Z(\omega)$, i.e., find C_n if $Z \overset{F.S.}{\longleftrightarrow} C_n$.

<u>Remark</u>: If appropriate convergence conditions are satisfied, observe the following:

(i) Since Z is periodic, the Fourier series coefficients c_n, $n \in \mathbb{Z}$, completely determine $Z(\omega)$ for all ω in the window $-\frac{\omega_0}{2} < \omega \leq \frac{\omega_0}{2}$.

(ii) The values of $Z(\omega)$ in the window

$$-\frac{\omega_0}{2} < \omega \leq \frac{\omega_0}{2}$$

completely determines $X(\omega)$, and hence, completely determines $x(t)$ for all t.

$$c_n \xrightarrow{\text{Fourier series inversion}} Z(\omega) \xrightarrow{\text{since } \omega_{max} < \frac{\omega_0}{2}} X(\omega) \xrightarrow{\text{Fourier inversion}} x(t)$$

$Z(\omega)$ can be found from c_n, $n \in \mathbb{Z}$

$X(\omega)$ can be found from $Z(\omega)$

x can be found from X.

Thus, c_n, $n \in \mathbb{Z}$, completely determine $x(t)$ for all t.

Hence, we have that: if $\omega_{max} < \frac{\omega_0}{2}$ then the samples $x(nT_0)$, $n \in \mathbb{Z}$, contain all the information to reconstruct $x(t)$ at all t.

(3) Formula for $\sum_{n \in \mathbb{Z}} \frac{1}{n^2+a^2}$ where $a>0$ is real:

(a) Apply the Poisson summation formula to the signal $\frac{1}{t^2+a^2}$ with $T_0 = 1$.

(b) Use this to relate $\sum_{n=-\infty}^{+\infty} \frac{1}{n^2+a^2}$ to hyperbolic cotangent

$$\coth(x) = \frac{e^{x}+e^{-x}}{e^{x}-e^{-x}}.$$